

1. Administrative & Study Identification

Full Study Title: Pooled Analysis of Single-arm Studies of Budesonide/Glycopyrronium/Formoterol (BGF) in Routine Care Setting **Short Title/Acronym:** CHOROS **Protocol Number:** D5980R00084 **NCT Number:** NCT06712563 **Posting ID:** 571258379359771024 **Sponsor Name:** AstraZeneca **Company:** AstraZeneca **Investigational Product:** Budesonide/glycopyrrolate/formoterol fumarate (BGF) **Phase of Development:** Not Applicable (Observational) **Trial Status:** RECRUITING (Currently recruiting participants) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To describe patients' clinical and self-reported outcomes of treatment with BGF in routine care settings through effectiveness, clinical, and self-reported measures assessed pre- and post-treatment initiation. **Secondary Objectives:** To assess changes in the COPD Assessment Test (CAT), Patient Global Impression of Severity (PGI-S) score, St. George's Respiratory Questionnaire (SGRQ), COPD Assessment Test Responder Rate, and exacerbation rates over up to 52 weeks of observation.

2.2 Background & Rationale The availability of single-inhaler triple therapies (such as BGF) in routine care may improve symptoms control, reduce exacerbation rates, and lower COPD-related healthcare costs, leading to an improvement in patient health status and quality of life. However, real-world studies focusing on Patient-Reported Outcomes (PROs) are still limited. The CHOROS study addresses this gap by retrospectively pooling integrated individual participant data from a series of planned and on-going primary prospective, non-interventional, multi-center real-world studies (e.g., CHOROS ORION, EBISU) conducted in the pulmonary/primary care practitioner setting in multiple countries (including the United Kingdom, Germany, Italy, Greece, Japan, Canada, and Romania) to evaluate short- and long-term impacts of BGF in a broader population.

3. Study Design & Methodology

3.1 Design Framework Study Type: OBSERVATIONAL (Retrospective secondary data use analysis of integrated individual participant data) **Control Type:** Single-arm (BGF Initiators) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Targeted Enrollment: 2,000
Number of Sites: Multiple sites across the United Kingdom, Germany, Italy, Greece, Japan, Canada, and Romania **Estimated Study Start:** 17-Apr-2025 **Estimated Primary Completion:** 14-Dec-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Patients with moderate to severe COPD and initiating BGF treatment according to the local prescribing guidance and as per their treating physician's recommendation. 3. Enrolled in on-going, single-arm, non-interventional studies sponsored by AstraZeneca. 4. Provided consent for secondary use of data.

4.2 Exclusion Criteria 1. Did not provide consent for the secondary use of data. 2. Patients hospitalized due to acute COPD exacerbations in the 30 days prior to enrollment (based on primary study protocols).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Drug Name: Budesonide/Glycopyrronium/Formoterol (BGF) **Dosage/Frequency:** Administered according to local prescribing guidance. **Route of Administration:** Oral inhalation (Pressurized Metered-Dose Inhaler / pMDI). **Duration of Treatment:** Up to 52 weeks of observation. **Budesonide ATC Codes:** R03BA02, R01AD05, A07EA06, D07AC09 **Budesonide Trade Names:** Symbicort, Rhinocort Aqua, Entocort, Preferid, Pulmicort, Rhinocort, Airsupra, Eohilia, Breyna, Tarpeyo, Breztri, Ortikos, Uceris

5.2 Concomitant & Prohibited Therapy Permitted: Medications for COPD and associated comorbidities prescribed as standard routine care. **Prohibited:** N/A (Observational standard-of-care setting).

6. Study Timeline & Visit Schedule

Note: As a retrospective pooled analysis of real-world data, the exact visit schedule corresponds to the primary study visits which generally follow standard clinical practice intervals.

Visit ID	Window	Key Activities/Procedures
Baseline	Day 0	Patient enrollment, demographics, medical history, baseline CAT and PGI-S scores, prior exacerbation history.
Interim 1	Week 12	CAT, PGI-S score evaluation, exacerbation rate.
Interim 2	Week 26	Clinical and self-reported outcomes assessment.
End of Study	Week 52	Final outcomes evaluation, CAT, PGI-S, annualized exacerbation rate.

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **COPD Assessment Test (CAT):** Compare self-reported COPD health status measured by COPD Assessment Test (CAT) at 12 weeks/3 months as compared to baseline in patients initiating treatment with BGF. 2. Change in annualized exacerbation rate in the first 12 weeks of treatment.

7.2 Secondary & Exploratory Outcomes 1. COPD Assessment Test (measured at subsequent follow-up intervals). 2. COPD Assessment Test Responder Rate. 3. Annualized rate of moderate or severe COPD exacerbations up to 52 weeks. 4. Change in Patient Global Impression of Severity (PGI-S) scores. 5. Change in St. George's Respiratory Questionnaire (SGRQ) total score.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine clinical practice monitoring; events collected through standard healthcare encounters. **Serious Adverse Events (SAEs):** Pharmacovigilance reporting as required for marketed products during routine care. **Data Monitoring Committee (DMC):** N/A (Retrospective pooled analysis).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All enrolled patients with moderate to severe COPD who initiated BGF and provided consent (Targeted Enrollment = 2,000). **Sample Size Justification:** A pooled sample size of 2,000 enables robust, multi-national real-world evidence generation across diverse populations and healthcare settings. **Handling of Missing Data:** Standard handling for observational real-world evidence, adjusting for confounding factors present in primary care.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Clinical Practice (GCP), the Declaration of Helsinki, and applicable local privacy laws (e.g., GDPR) for secondary data use. **Monitoring Plan:** Retrospective pooling of secondary data via secure IQVIA servers. **Audit Trail:** De-identified patient data maintained using encrypted study registration numbers to protect patient privacy.

11. Study Identifiers & Governance

Primary ID: D5980R00084 **Registry Number:** NCT06712563 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** CHOROS **Official Regulatory**

Title: Pooled Analysis of Multi-country, Open-label Single-arm, Non-interventional, Multi-center, Cohort Studies of Real-world Outcomes in New Users of Budesonide/Glycopyrronium/Formoterol (BGF) in Routine Care Setting

12. Clinical Framework (COPD Specific)

Disease Area: Pulmonary Disease, Chronic Obstructive (Preferred Term: Lung diseases) **Semantic Type:** Disease or Syndrome **Condition Definition:** Pathological processes involving any part of the LUNG. **Diagnosis Threshold:** Moderate to severe COPD **MeSH Code:** D008171 **HPO Code:** HP:0002088 **SNOMED ID:** 19829001 **Medical Methodology:** Routine Care / Standard clinical pathway **Optimization Target:** Improvement in COPD symptoms, exacerbation reduction, and Patient-Reported Outcomes (PROs)

13. Study Procedures & Methodology

Management Pathway: Retrospective secondary data use analysis **Education Protocol (HCP):** N/A (Observational) **Education Protocol (Patient):** N/A (Observational) **Quality Audit Mechanism:** Secure data transfer and retrospective integration of multi-national EHR/PRO data

14. Observation & Follow-Up Schedule

Total Duration: Up to 52 Follow-up weeks **Onsite Visit Cadence:** Routine clinical practice visits (typically assessed at Weeks 12, 26, and 52) **Remote Monitoring Frequency:** N/A **Checklist Review Interval:** N/A

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				